

Derren Winata

[LinkedIn](#) | (+65) 8355 3698 | wderren17@gmail.com | [Website](#) | [GitHub](#)

NUS Data Science & Analytics, Class of 2026 — ships AI apps, data products, and full-stack software.

Education

National University of Singapore, Singapore, SG

Aug '22 – Aug '26

Bachelor of Science, Data Science and Analytics

Coursework: Data Structures and Algorithms, Probability, Data Analytics Tools, Linear Algebra, Calculus, Programming Methodology (Python)

Experience

AI Singapore, Singapore

Mar 2026 – Jul 2026

Programme & Partnerships Assistant · Part-time

- Built and deployed the AI for Good website (Next.js, TypeScript, Tailwind CSS) covering impact stats, SDG alignment, and ASEAN partnership footprint.
- Developed the NSWS automation platform (an internal operations tool) using FastAPI, PostgreSQL, Celery, ChromaDB, and LangGraph for AI-assisted operations, briefs, compliance reminders, and reporting agents.
- Built the AI Ready ASEAN Youth Challenge 2026 Judging Portal (Next.js, Supabase) with judge assignments, weighted scoring, dashboards, CSV export, and audit logs.

360 Cogni, Singapore

Jan 2026 – Jul 2026

Product Manager Intern

- Defined MVP scope and roadmap across cognitive screening, personalised results, and brain training resources for 1,000+ users.
- Designed 5+ key flows (onboarding, demographics, assessment, results, resources) with mobile-first UX across iOS, Android, and desktop.
- Specified integrations (GMS assessment API, Supabase, Vercel, SendGrid), database schema, and Edge Functions for email automation; tracked conversion funnels and folded 20+ user-testing improvements into the shipped MVP at 360cogni.com.

AI Singapore, Singapore (Remote)

Mar 2024 – Jul 2026

Quality Assurance Assistant · Part-time

- Validated datasets used in product launches, reducing post-launch errors by 30%.
- Cross-referenced 1,000+ data files against official government and third-party sources.

Sunnystep, Singapore

Mar 2026 – Apr 2026

AI Agent Intern

- Built a Node.js marketing intelligence pipeline with Anthropic Claude for daily briefs, monthly calendars, and weekly performance views.
- Automated delivery via scheduler so stakeholders received structured updates without manual prep.

National University of Singapore, Singapore

Dec 2025 – Mar 2026

Full-Stack Web Developer — Educational Jigsaw Game · Part-time

- Built lobby, shareable sessions, Game Master controls, two-phase puzzles, timers, points, and post-round results.
- Delivered touch/drag puzzle play on mobile and used Supabase for real-time updates and leaderboards.

National University of Singapore, Singapore

Nov 2025 – Feb 2026

Student Researcher · Part-time

- Architected a React/TypeScript and Node.js/Express app for EQ-5D-5L Classic TTO health economics research.
- Built real-time utility calculations, Admin/Interviewer dashboards, and an AI co-pilot for protocol guidance.
- Implemented GDPR-aligned auth and encryption, plus WCAG 2.1 AA support in English, Spanish, Chinese, and Indonesian.

National University of Singapore, Singapore

Sep 2025 – Nov 2025

Web & Mobile Development Intern · Part-time

- Developed a full-stack web and mobile app for cognitive health screening in DSFP, a national cognitive-health screening programme.
- Added accessibility, English/Chinese/Malay support, and a caregiver management view with analytics.

National University of Singapore, Singapore

Sep 2025 – Oct 2025

Data Engineering Research Assistant · Part-time

- Built an end-to-end pipeline scraping, cleaning, and validating 2,695+ halal-certified establishments from MUIS, Singapore's Islamic religious authority.
- Reached 99.72% address coverage and 100% postal code coverage across 28 postal districts.
- Shipped 30+ Python scripts, quality reports, and reproducible Makefile workflows.

Firsty.app, Singapore (Hybrid)

May 2025 – Jul 2025

Client Development Intern

- Designed and automated a LinkedIn outreach workflow for prospect filtering and message sequencing.
- Identified 300+ APAC companies based on ICP criteria to sharpen targeting.

Medisaya, Singapore (Hybrid)

Feb 2025 – Jun 2025

Software Developer

- Developed Python scrapers that extracted 50k+ records and cut manual collection time by 70%.
- Built and maintained Django REST APIs for secure health data access.

Marina Bay Sands, Singapore (Hybrid)

May 2024 – Dec 2024

eCommerce Analytics Intern

- Automated a critical stakeholder report with Python and Power Automate, reducing preparation time by ~83%.
- Led the website Data Layer rollout, improving overall data accuracy to 98%, and configured Tealium IQ tracking, increasing reliability of captured data by 20%.
- Analysed traffic and marketing data with recommendations that lifted conversion rates by 15%, and partnered with Performance Marketing on A/B tests that improved campaign ROI by 10%.

Selected Projects

360 Cogni — React 19, React Native, Supabase · dementia and cognitive health platform, live at 360cogni.com (1,000+ user target)

EQ-5D-5L TTO Research Tool — React, Node.js, PostgreSQL · real-time health-economics utility scoring with an AI co-pilot, WCAG 2.1 AA and GDPR-aligned

Bayesian Pair Trading — Python, Optuna · walk-forward optimised S&P 500 pairs trading with dual cointegration testing

Pulse — Social Intelligence — Go, PostgreSQL, Kafka · concurrent Hacker News and Reddit crawler streaming scored events through Kafka

Sunnystep Strides — React 18, Supabase, Recharts · AI marketing automation dashboard with agent-driven briefs and content calendars

Halal Food Landscape — Python, Pandas, Selenium · MUIS registry pipeline covering 2,695+ establishments at 99.72% address coverage

Skills

Languages: Python, TypeScript, SQL, Go, Bash/Shell scripting

Frontend: React 19, React Native, Next.js, Expo, Tailwind CSS, Recharts

Backend & APIs: Node.js, Express, FastAPI, Django REST Framework, Supabase, Vercel, Edge Functions

Data & Pipelines: PostgreSQL, Snowflake, Databricks, ChromaDB, Pandas, Apache Airflow, Kafka, Celery, ETL/ELT, data warehousing, database administration, RBAC

AI & LLM: LangGraph, Anthropic Claude, Amazon Bedrock, Model Context Protocol (MCP), RAG, prompt engineering, tool calling, agent monitoring

Analytics & Measurement: Tealium IQ, Google Analytics, Tableau, Power Automate, Selenium, Optuna, A/B testing, conversion funnels, KPI optimisation

Product: MVP scoping, roadmap definition, UX flows, user testing, stakeholder reporting

Accessibility & Compliance: WCAG 2.1 AA, GDPR-aligned auth and encryption

Practices: Reproducible Makefile workflows, quality reporting, code review, QA sign-off

Certifications

AI & LLM: Snowflake Generative AI Professional Certificate — Snowflake · Coursera, Sep 2026 (ID M5ZEUB06DKRU); Claude with the Anthropic API, Claude with Amazon Bedrock, Claude Platform 101, AI Fluency: Framework & Foundations, Introduction to Model Context Protocol — Anthropic, Aug 2026; AI Fundamentals: Language and Vision in AI, AI Fundamentals: Foundations for Understanding AI — IBM, Aug 2026

Data & Analytics: IBM Relational Database Administrator with GenAI Professional Certificate — IBM · Coursera, Sep 2026 (ID IGHUSCNY8XUT); Snowflake Data Engineering Professional Certificate — Snowflake · Coursera, Sep 2026 (ID 4GUSO181FPT6); ETL and Data Pipelines with Shell, Airflow and Kafka — IBM · Coursera, Sep 2026 (ID LBCATH579UE3); Google Data Analytics Professional Certificate — Google · Coursera, Aug 2026 (ID ZLCROBCFKI5Y); Databricks Fundamentals Accreditation — Databricks, Aug 2026; Google Analytics Certification — Google, Aug 2026 (expires Aug 2027, ID 191811640); Data Analytics and Visualization Job Simulation — Accenture · Forage, Dec 2023

Project Management: Google Project Management Professional Certificate — Google · Coursera, Aug 2026 (ID UBNQ12MMPGA0)

Finance: Bloomberg Market Concepts (BMC) — Bloomberg for Education, Sep 2026

Awards & Languages

Awards: Bronze Award — Mathematical Olympiads. Thailand IMO & GBA Olympiad • 2021–2022

Languages: Fluent in English and Bahasa Indonesia; limited working proficiency in Chinese